TOP 5 LOW-LEVEL DESIGN GAMES

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Most Asked LLD Interview Questions

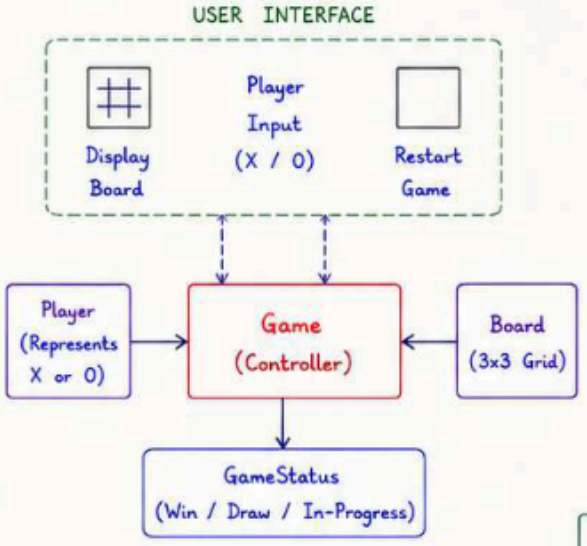
1. Tic Tac Toe | 2. Snake & Ladder | 3. Chess | 4. Ludo | 5. UNO Card Game

1. TIC TAC TOE (LOW-LEVEL DESIGN)

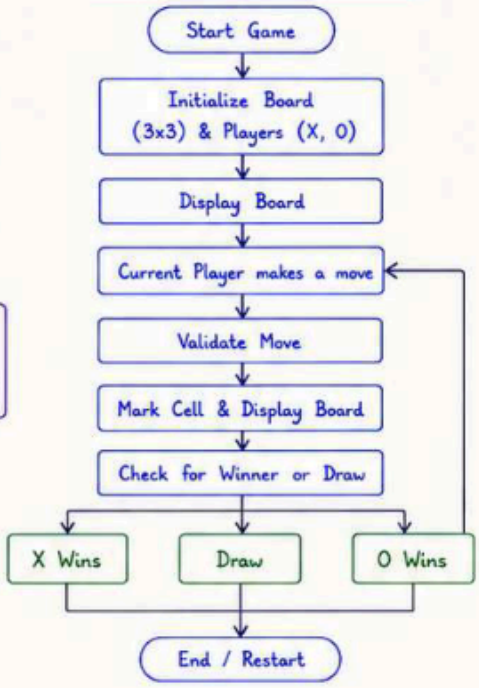
1. Requirements

- ✓ Two players : X and O
- ✓ 3x3 Board
- ✓ Players take turns
- ✓ Valid moves only
- ✓ Detect winner or draw
- ✓ Display board after each move
- ✓ Restart game

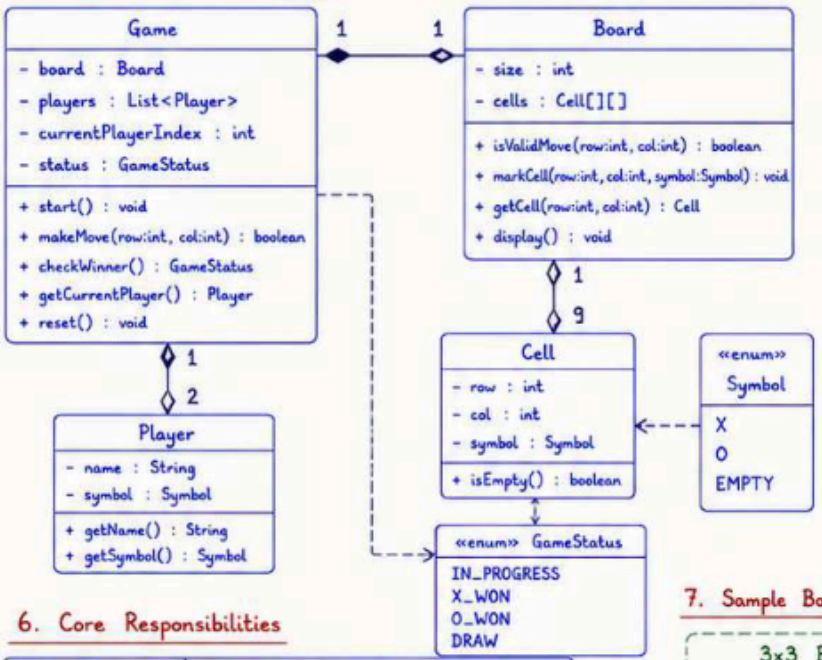
2. System Design Diagram



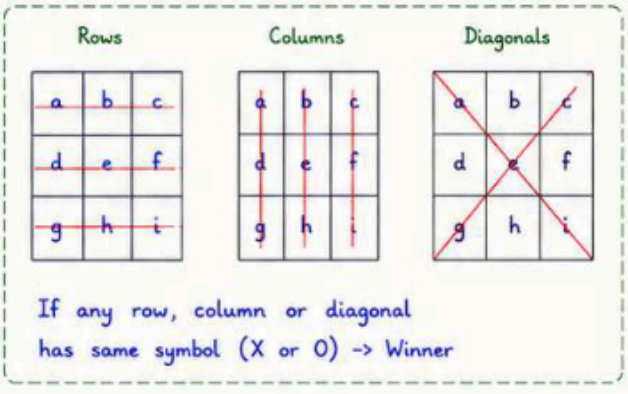
3. Game Flow (High Level)



4. High Level Class Diagram



5. Winner Checking Logic



6. Core Responsibilities

Class	Responsibility
Game	Controls game flow, manages turns, checks winner and maintains game status.
Board	Maintains 3x3 grid, validates moves and updates cells.
Player	Represents a player with name and symbol (X/O).
Cell	Represents a single cell in the board.
Symbol (Enum)	Represents X, O and EMPTY values.
GameStatus (Enum)	Represents current status of the game.

7. Sample Board Representation

3x3 Board

X	O	X
O	X	
	O	

Or as 2D Array ->

```
Cell[][] cells = {
  {X, O, X},
  {O, X, EMPTY},
  {EMPTY, O, EMPTY}
};
```

8. Design Pattern Used (Optional)

Strategy Pattern (for Winner Checking)

- Different strategies for checking winner : Row, Column, Diagonal.
- Makes code extensible and maintainable.

9. Key Points

- ★ Only EMPTY cells can be marked.
- ★ After each move, check for winner.
- ★ If all cells are filled and no winner -> DRAW.
- ★ Players take turns.
- ★ Game can be reset anytime.

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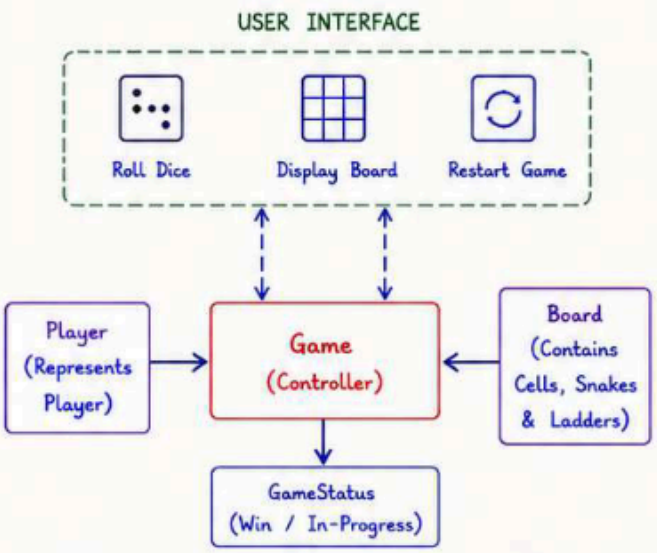
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2. SNAKE & LADDER (LOW-LEVEL DESIGN)

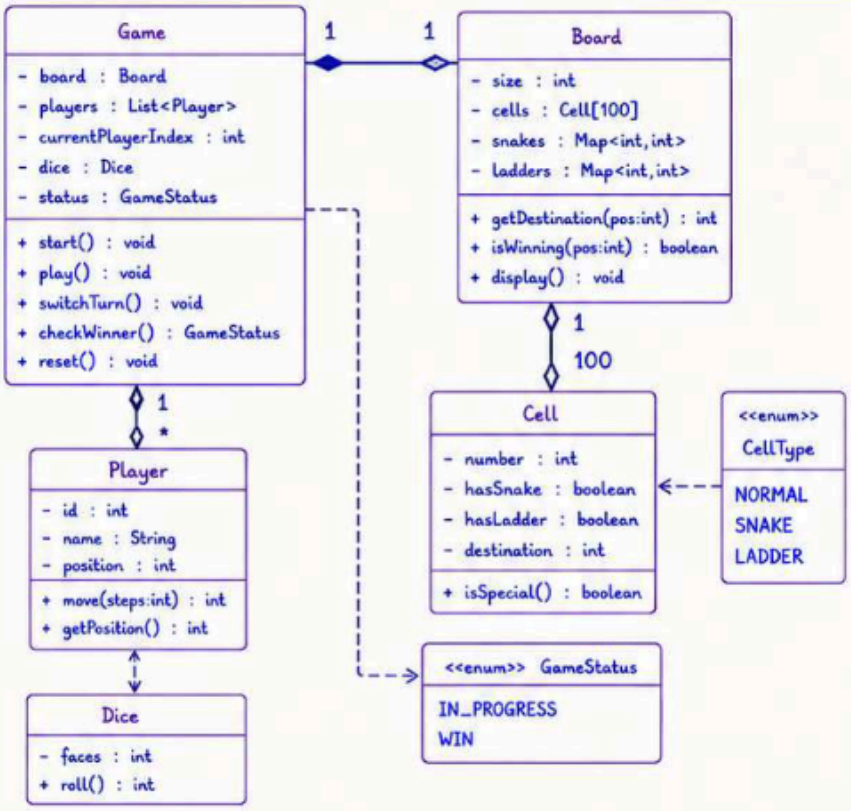
1. Requirements

- ✓ 2 to N players.
- ✓ Board with 100 cells (1 to 100).
- ✓ Players take turns to roll dice.
- ✓ Snake moves player down.
- ✓ Ladder moves player up.
- ✓ First player to reach 100 wins.
- ✓ Exact roll required to win.
- ✓ Display current positions.

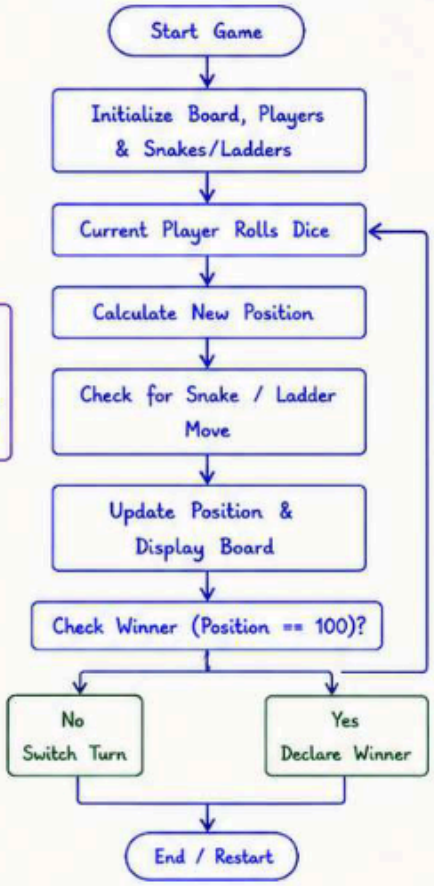
2. System Design Diagram



3. High Level Class Diagram



4. Game Flow (High Level)



6. Core Responsibilities

Class	Responsibility
Game	Controls game flow, manages turns, checks winner and maintains status.
Board	Maintains board, snakes, ladders and validates moves.
Player	Represents a player and tracks current position.
Cell	Represents a single cell and its type (snake / ladder / normal).
Dice	Generates random number on roll.
GameStatus	Represents current status of the game.

7. Key Points

- ★ Player follows board path after dice roll.
- ★ If cell has ladder → move up.
- ★ If cell has snake → move down.
- ★ Exact roll needed to reach 100.
- ★ Game ends when a player reaches 100.

8. Extension Ideas

- ✓ Variable board size (50, 100, 150 etc.).
- ✓ Multiple dice (advanced version).
- ✓ Computer opponent.
- ✓ Save & resume game.
- ✓ Online multiplayer.

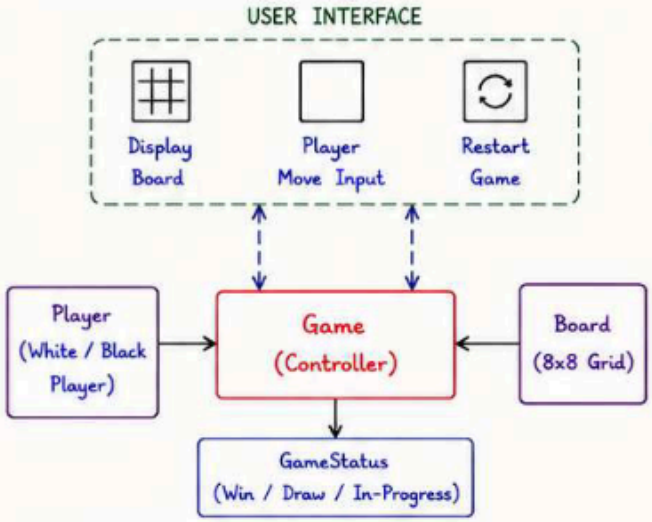
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3. CHESS (LOW-LEVEL DESIGN)

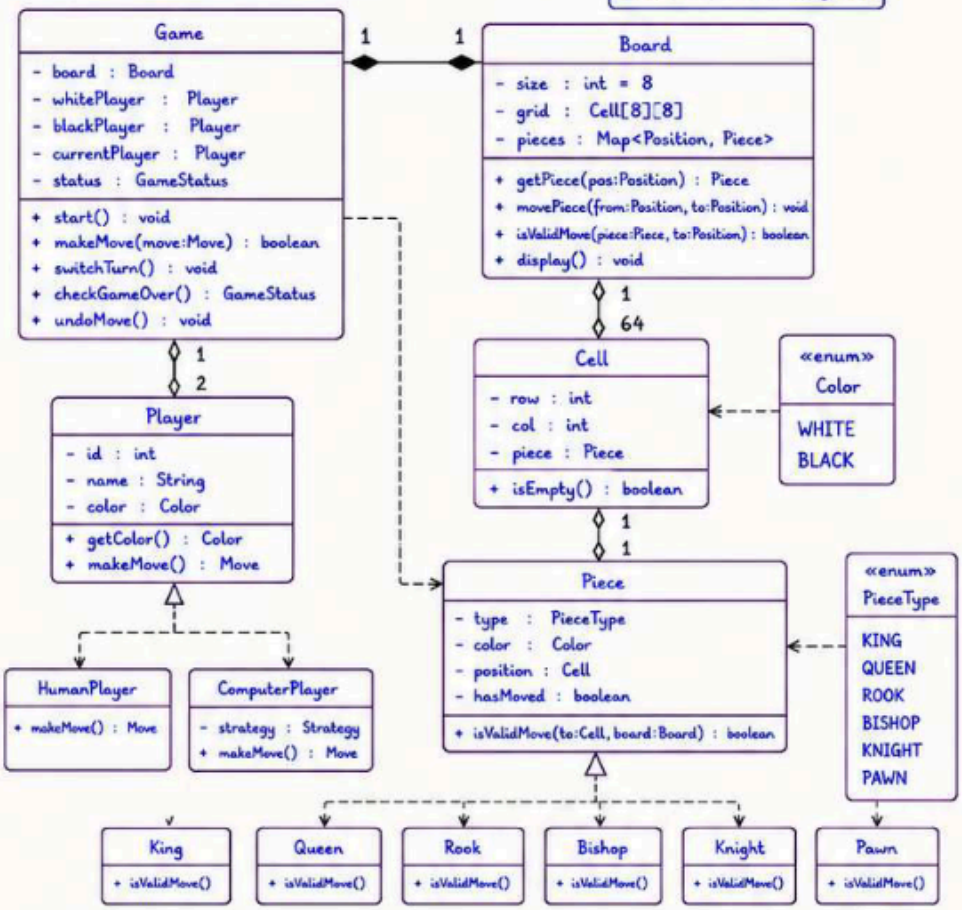
1. Requirements

- ✓ Two players.
- ✓ Standard 8x8 chess board.
- ✓ All chess pieces with valid moves.
- ✓ Players take turns.
- ✓ Detect check, checkmate, stalemate.
- ✓ Game ends when a player wins.
- ✓ Display current board state.
- ✓ Support undo move (optional).

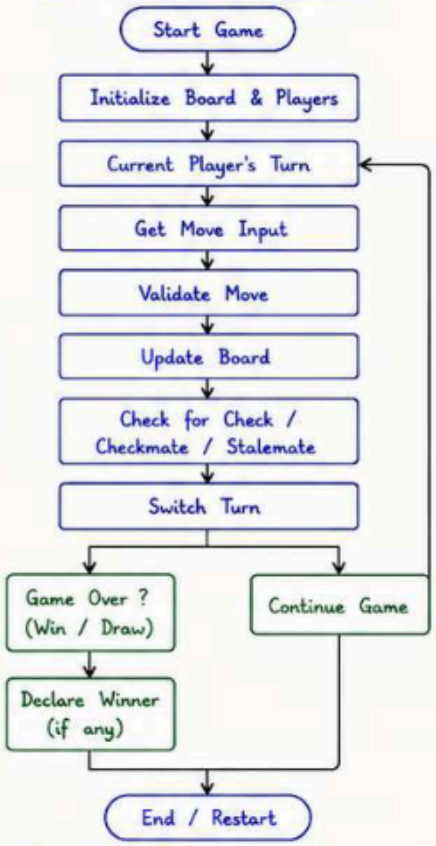
2. System Design Diagram



3. High Level Class Diagram



4. Game Flow (High Level)



5. Core Responsibilities

Class	Responsibility
Game	Controls game flow, manages turns and checks game status.
Board	Maintains 8x8 grid, pieces and validates moves.
Player	Represents a player and provides the color.
Piece	Abstract class for all pieces. Contains movement rules.
Cell	Represents a single cell on (White/Black).
Color (Enum)	Represents type of piece.
GameStatus (Enum)	Represents current status of the game.

6. Key Points

- ★ Each piece has its own movement rules.
- ★ Validate move before updating board.
- ★ Check for check, checkmate and stalemate.
- ★ Switch turn only after a valid move.
- ★ Game ends on checkmate or stalemate.
- ★ Undo/redo can be added as an extension.

7. Extension Ideas

- ✓ AI player with different difficulty levels.
- ✓ Timer for each player.
- ✓ Undo / Redo moves.
- ✓ Save and load game.
- ✓ Online multiplayer support.

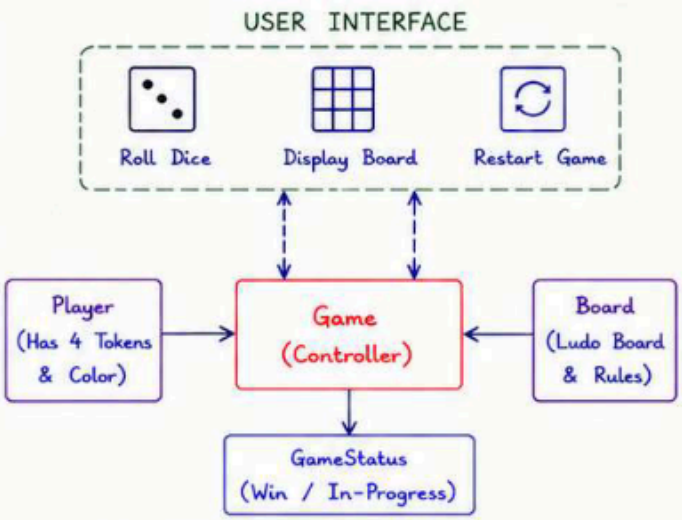
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4. LUDO (LOW-LEVEL DESIGN)

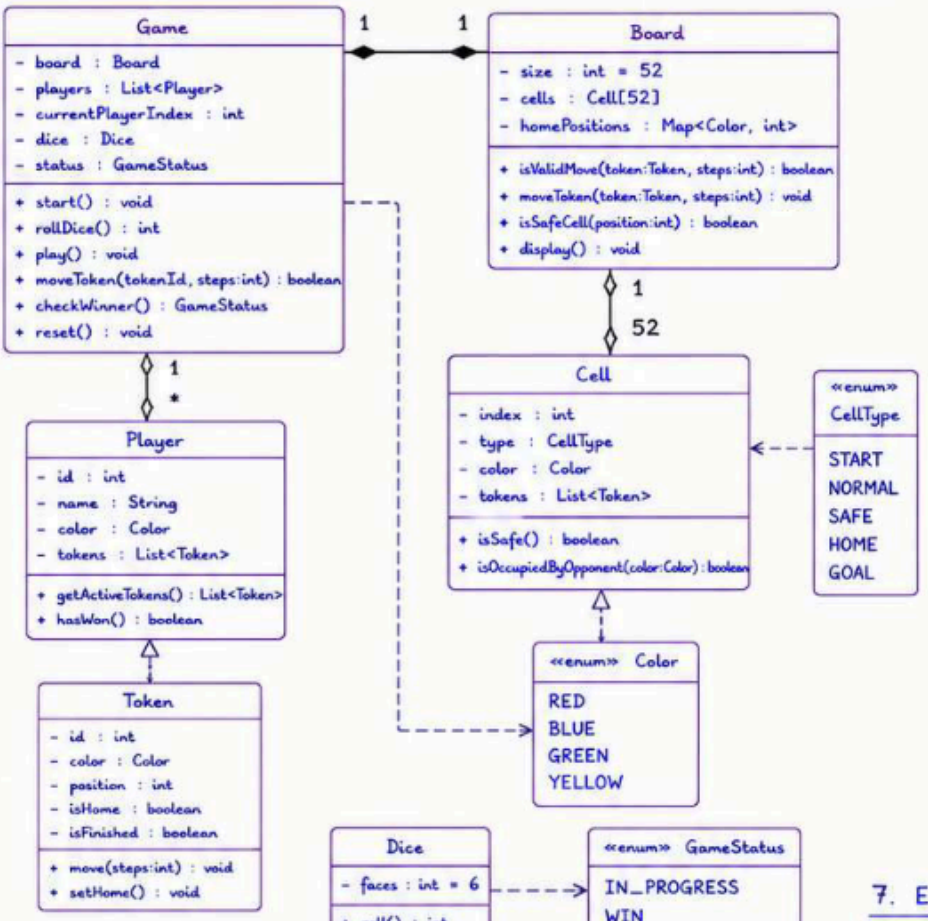
1. Requirements

- ✓ 2 to 4 players.
- ✓ Standard Ludo board.
- ✓ Each player has 4 tokens.
- ✓ Players take turns to roll dice.
- ✓ Token moves based on dice value.
- ✓ First player to get all tokens to home wins.
- ✓ Display board & current status.

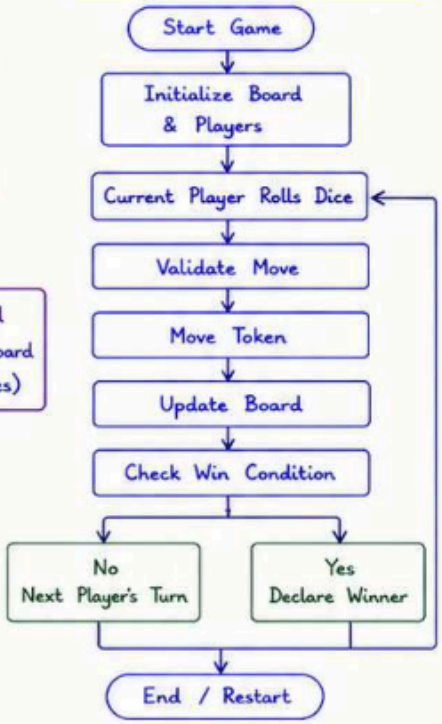
2. System Design Diagram



3. High Level Class Diagram



4. Game Flow (High Level)



5. Core Responsibilities

Class	Responsibility
Game	Controls game flow, manages turns, dice roll and win condition.
Board	Maintains Ludo board, validates moves, updates token positions.
Player	Represents a player and manages their tokens.
Token	Represents a token and its current position.
Cell	Represents a cell on board and its type.
Dice	Generates random number on roll.
GameStatus	Represents current status of the game.
Color (Enum)	Defines player colors.
CellType (Enum)	Defines types of cells.

6. Key Points

- ★ Player must roll 6 to move a token out of START.
- ★ A token can only move exact dice steps.
- ★ If token lands on opponent, opponent token goes back to START.
- ★ Safe cells cannot capture tokens.
- ★ All 4 tokens must reach HOME to win.
- ★ Players take turns in clockwise order.

7. Extension Ideas

- ✓ Support 2, 3 or 4 players.
- ✓ AI player to play automatically.
- ✓ Chat / Voice while playing.
- ✓ Save & resume game.
- ✓ Online multiplayer support.
- ✓ Multiple board themes.

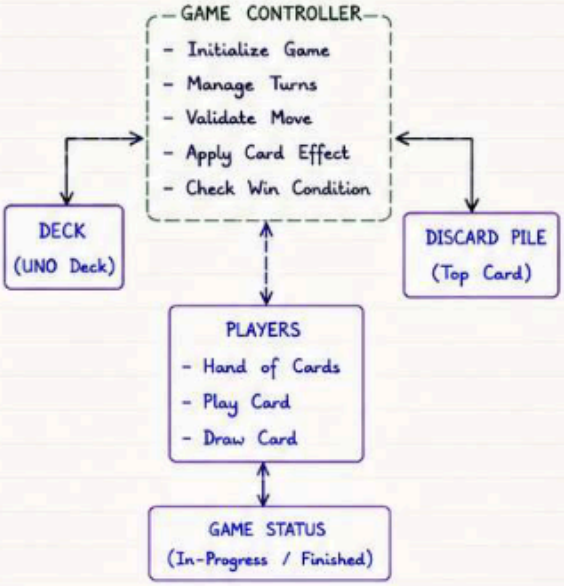
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5. UNO CARD GAME (LOW-LEVEL DESIGN)

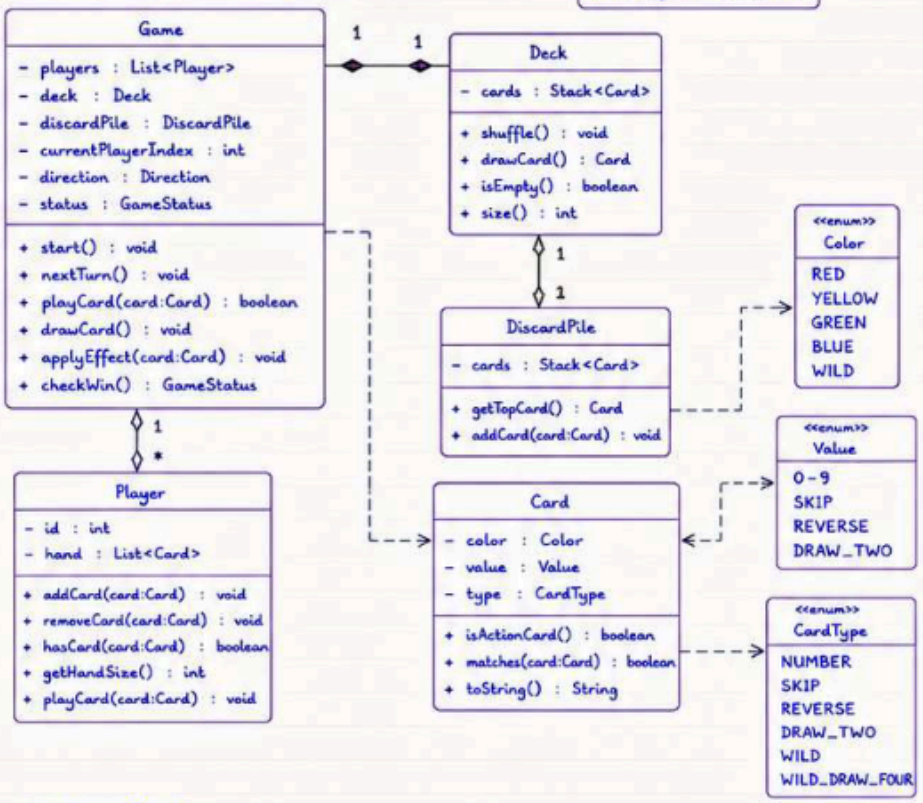
1. Requirements

- ✓ 2 to 10 players.
- ✓ Standard UNO deck (108 cards).
- ✓ Each player has 7 cards.
- ✓ Players take turns to play a card.
- ✓ A card must match color, number or symbol.
- ✓ Special action cards have specific effects.
- ✓ First player to play all cards wins.
- ✓ Display current card and direction.

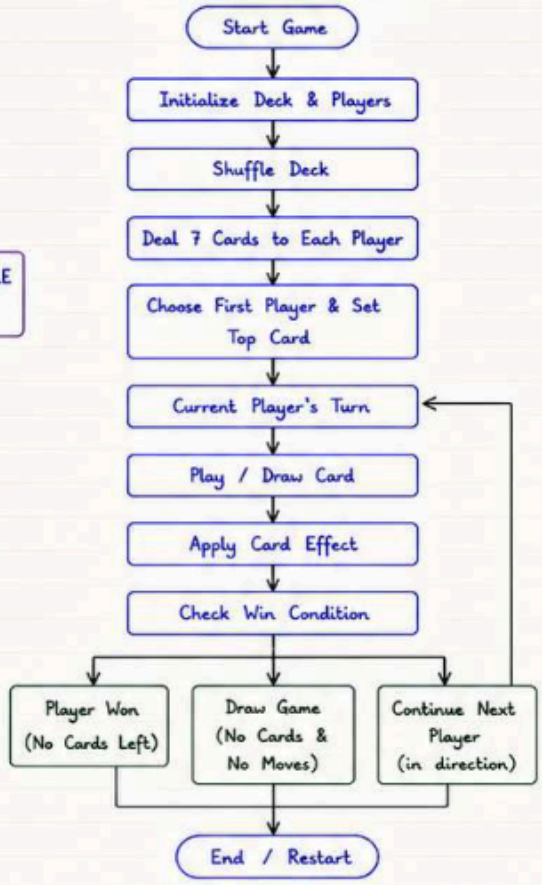
2. System Design Diagram



3. High Level Class Diagram



4. Game Flow (High Level)



5. Core Responsibilities

Class	Responsibility
Game	Controls game flow, manages turns, applies effects and checks win.
Deck	Holds UNO deck, shuffles and provides cards.
DiscardPile	Maintains discard pile and top card.
Player	Represents a player and manages hand of cards.
Card	Represents a UNO card with color, value and type.
GameStatus	Holds current status of the game.
Direction	Tracks play direction (Clockwise or Counter-Clockwise).

6. Key Points

- ★ A card can be played if color, number or type matches.
- ★ Action cards have immediate effects.
- ★ Wild cards can be played on any card.
- ★ Draw Two → next player draws 2 cards & skips turn.
- ★ Reverse → changes the direction of play.
- ★ Wild Draw Four → next player draws 4 cards & skips turn.
- ★ First player to have 0 cards wins.

7. Extension Ideas

- ✓ AI players with different difficulty levels.
- ✓ Custom rules (stacking draw two, etc.).
- ✓ Score calculation system.
- ✓ Online multiplayer support.
- ✓ Add timer for each player's turn.
- ✓ Chat / Voice while playing.
- ✓ Different UNO variations (7-0, Flip, etc.).